

LAANE-NIGULA, Estonia

WMO: 261240

Lat: 58.9511N

Lon: 23.8156E

Elev: 25

StdP: 101.02

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.8	-15.9	-22.2	0.5	-19.5	-18.7	0.7	-15.4	10.2	1.1	9.4	0.7	1.9	100	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	27.6	19.4	25.5	18.3	23.7	17.5	20.7	25.4	19.5	23.9	18.5	22.4	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.9	17.9	12.9	21.3	16.9	12.1	20.4	59.6	25.5	55.7	24.0	52.3	22.5	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.2	7.3	-22.7	29.8	4.6	2.8	-26.0	31.9	-28.7	33.5	-31.3	35.1	-34.7	37.2		
(5)			-23.2	22.1	4.1	1.4	-26.2	23.2	-28.6	24.0	-30.9	24.8	-33.9	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-4.0	-4.0	-0.9	5.1	10.7	14.1	17.6	16.4	11.7	6.4	2.5	-0.3		
	DBStd	8.53	5.93	5.69	4.46	3.65	4.16	3.14	3.05	2.90	3.05	3.99	3.95	4.95		
	HDD10.0	2046	433	392	339	153	39	3	0	1	18	123	225	321		
	HDD18.3	4430	692	625	597	398	238	132	51	75	198	371	475	579		
	CDD10.0	708	0	0	0	6	62	127	236	198	70	10	0	0		
	CDD18.3	52	0	0	0	0	3	6	28	14	0	0	0	0		
	CDH23.3	481	0	0	0	0	40	68	256	116	1	0	0	0		
	CDH26.7	93	0	0	0	0	4	8	60	22	0	0	0	0		
	WSAvg	3.5	3.8	3.8	3.7	3.5	3.3	3.4	2.9	3.0	3.1	3.4	3.8	4.3		
	PrecAvg	663	54	40	35	34	37	59	70	79	58	72	67	59		
	PrecMax	966	111	103	67	72	74	105	133	170	119	136	131	159		
(15) Precipitation	PrecMin	375	10	6	4	5	5	12	7	7	9	14	2	16		
	PrecStd	119	24	21	17	20	18	25	33	36	29	32	31	28		
	DB	6.2	5.3	13.0	20.9	26.8	27.7	30.5	29.2	22.0	16.4	10.2	9.0			
	MCWB	5.6	4.0	6.5	12.3	17.3	18.9	20.6	20.4	16.1	13.1	9.1	8.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	4.7	4.1	8.5	17.6	24.2	25.3	28.4	26.3	19.9	14.2	9.2	7.5			
	2% MCWB	4.1	3.1	5.0	10.1	15.9	17.9	20.1	19.0	15.4	12.3	8.3	6.5			
	5% DB	3.4	3.1	5.8	14.8	21.6	22.9	26.2	24.1	18.4	13.0	8.5	6.5			
	5% MCWB	2.8	2.4	3.5	8.5	14.2	16.3	19.0	17.9	14.6	11.7	7.7	5.7			
	10% DB	2.5	2.3	4.2	12.4	18.9	21.0	24.2	22.3	17.0	11.9	7.6	5.3			
	10% MCWB	2.0	1.7	2.5	7.4	13.2	15.1	18.3	17.2	14.1	10.8	6.9	4.7			
	0.4% WB	5.7	4.4	7.6	13.2	19.4	20.8	22.3	21.7	17.3	13.7	9.5	8.1			
	0.4% MCDB	6.0	4.9	11.4	19.3	23.6	25.6	27.6	26.5	20.0	14.8	9.9	8.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	4.2	3.3	5.3	10.6	17.2	18.7	21.1	20.0	16.2	12.8	8.6	6.9			
	2% MCDB	4.4	3.9	7.8	15.7	22.2	23.3	26.1	24.5	18.8	13.9	9.0	7.4			
	5% WB	2.9	2.5	3.8	9.3	15.5	17.2	20.0	18.9	15.3	11.9	7.8	5.9			
	5% MCDB	3.3	3.1	5.5	14.1	20.1	21.9	24.7	22.9	17.6	12.8	8.3	6.4			
	10% WB	2.0	1.7	2.8	7.9	13.5	15.9	19.0	17.9	14.5	11.0	7.0	4.7			
	10% MCDB	2.4	2.2	4.0	11.6	17.8	20.0	23.5	21.1	16.5	11.8	7.6	5.3			
	MDBR	4.7	5.2	6.8	9.3	10.7	9.6	9.4	9.6	8.3	6.2	3.8	3.9			
	5% DB	3.7	4.1	8.9	14.8	14.5	13.4	12.9	12.3	10.5	7.0	4.0	3.4			
(36) Mean Daily Temperature Range	5% WB	3.8	3.8	6.0	8.4	7.3	6.8	5.7	6.2	6.4	5.1	3.7	3.3			
	MCDBR	3.8	3.9	8.0	13.2	12.2	11.4	11.3	10.5	8.7	5.8	3.7	3.0			
	MCWBR	3.8	3.7	5.7	7.9	6.7	6.4	5.5	5.9	5.8	4.6	3.7	3.2			
	MCWBR	3.8	3.7	5.7	7.9	6.7	6.4	5.5	5.9	5.8	4.6	3.7	3.2			
(40) Clear-Sky Solar Irradiance	taub	0.282	0.310	0.334	0.372	0.366	0.364	0.392	0.386	0.354	0.336	0.317	0.274			
	taud	2.156	2.195	2.260	2.318	2.404	2.439	2.382	2.395	2.469	2.407	2.228	2.119			
	Ebn at Noon	517	695	799	831	865	872	833	810	781	673	469	387			
	Edh at Noon	47	75	95	107	105	103	107	99	78	62	45	35			
(44) All-Sky Solar Radiation	RadAvg	0.28	0.84	2.36	3.99	5.60	5.88	5.54	4.36	2.70	1.20	0.34	0.18			
	RadStd	0.05	0.15	0.27	0.44	0.54	0.35	0.45	0.41	0.32	0.15	0.06	0.03			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (51 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-133	N/A	N/A	

Nomenclature: See separate page